What's new in Forecast Agent

Everything added to the project during development, and where to find it

Change summary · MDM Hackathon · 17 October 2026

The project started as a machine-learning module only: two forecasting models (an LSTM and an MLP), data validation, command-line tools to train, evaluate and forecast, and a synthetic demo building. It now links the two models into an ensemble, produces one structured forecast report, has a web app and a question-answering assistant, runs on a real office building, and is documented for judges and team.

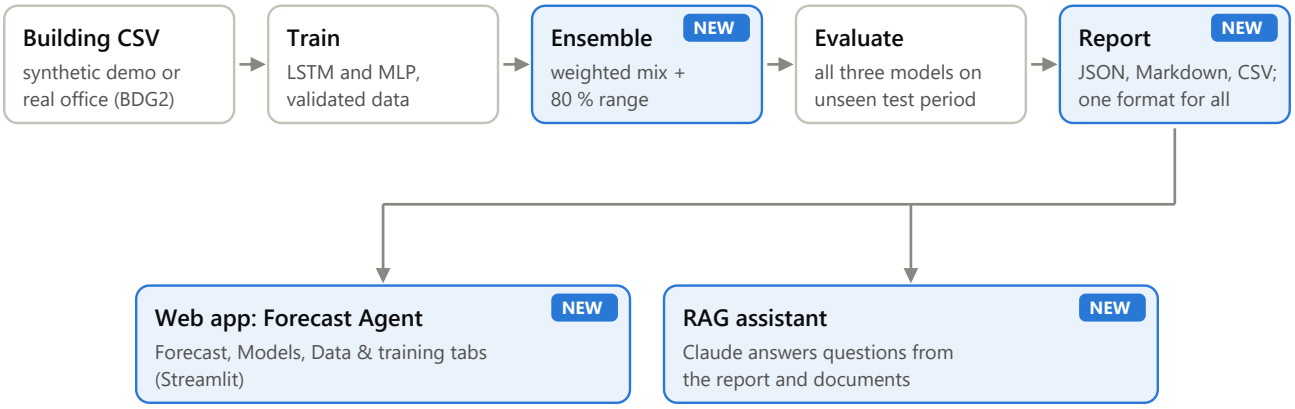
51 → 82 automated tests, all passing	**3** forecasting models: LSTM, MLP and the new ensemble	**3** web app tabs: Forecast, Models, Data & training	**1** real building dataset, ready to train on

1 Before and now

Area	Before	Now
Models	LSTM and MLP, compared side by side	Plus a **weighted ensemble** of the two with an 80 % forecast range for every hour
Forecast output	One table per forecast (CSV / JSON)	Plus a **forecast report** in one format for every model: summary, accuracy, range, hourly values, caveats (JSON and Markdown)
Interface	Command line only	Plus the **Forecast Agent web app** (Streamlit)
Questions in plain language	Not possible	**RAG assistant**: Claude answers from the forecast and the project documents (terminal)
Data	Synthetic demo building	Plus a **real 10,352 m² office** from the Building Data Genome Project 2, and a converter for any building in it
Results	Synthetic data only	Trained on the real office: LSTM test MAE 7.81 kW, WAPE 8.6 %
Documentation	README	Plus README sections, a dataset guide with license credit, and three PDFs
Tests	51	82, including a headless run of the web app

2 How the pieces fit together

New stages are highlighted. Everything reads and writes the same files, so the command line, the web app and the assistant always agree.



3 New features

3.1 Ensemble that links the two models

`src/ensemble.py · src/evaluate.py · src/predict.py`

- Forecast = w × LSTM + (1 − w) × MLP, with w fitted on the validation period only.
- An 80 % range per hour ahead, from the spread of validation errors; its real coverage on the test period is measured and reported.
- Scored next to both models and selectable as the best model; refuses to run if a model was retrained after it was fitted.

3.2 Forecast report

`src/report.py`

- One documented structure for every model: model and weights, period, peak, lowest, energy, accuracy of that exact model version, range, hourly values and caveats.
- Saved as JSON and Markdown next to every forecast; predict prints it (--format report | table | json).

3.3 Web app: Forecast Agent

`streamlit_app.py · src/ui/charts.py`

- **Forecast tab:** pick a model, CSV and optional backtest time; summary tiles, chart with range and both models, table view, downloads.
- **Models tab:** comparison table, ensemble weights, error by lead time, evaluation plots, re-run evaluation.
- **Data & training tab:** upload, validate, and one button that trains, evaluates and forecasts.
- One colour per model in every chart, checked for colour-blind readability, in light and dark mode.

3.4 RAG assistant

`src/rag/ · knowledge/`

- Searches the project documents (TF-IDF) and always adds the latest forecast and accuracy figures.
- Claude answers only from that context, in a fixed structure: answer, key figures, confidence, caveats, sources.
- Runs from the terminal; --retrieve-only shows what would be sent without calling the API.

3.5 Real building data

`tools/convert_bdg2.py · data/bdg2_hog_office_lizzie.csv`

- Downloads the Building Data Genome Project 2 with checksum checks, lists office buildings with clean data, and converts one into the project's CSV format.
- Included: a real office, hourly for 2016–2017 (17,544 rows), with outdoor temperature and humidity.

4 Results on the real office

Test period 2017-09-13 to 2017-12-31, never seen during training or model selection.

Model	Test MAE (kW)	Test RMSE (kW)	Test WAPE	Daily peak error (kW)
LSTM (preferred)	7.81	10.53	8.6 %	6.92
MLP	9.38	12.31	10.3 %	10.16
Ensemble	7.81	10.53	8.6 %	6.92

- On this building the ensemble settled on 1.00 × LSTM + 0.00 × MLP, because adding the MLP made validation error worse, so it equals the LSTM here.
- On the synthetic demo building it chose 0.40 × LSTM + 0.60 × MLP and beat both models (test MAE 5.15 against 5.43 kW).
- The 80 % range held 86.3 % of actual test readings.

5 Tests and documentation

Added	Files	Covers
31 new tests	`tests/test_ensemble.py, test_rag.py, test_ui.py, test_convert_bdg2.py, test_workflow.py`	Ensemble maths and report format; retrieval, prompt and answer handling; charts and a headless app run; the dataset converter; ensemble forecasting end to end.
Guides	`README.md, data/README.md`	Ensemble, report schema, assistant, web app; dataset conversion and license credit.
PDFs	`docs/Forecast_Agent_Overview.pdf, docs/Forecast_Agent_RAG.pdf, this document`	Project overview, RAG implementation, change summary.
Setup	`requirements.txt, .venv/`	New dependencies: anthropic, pypdf, streamlit.

6 Changes made on request

- The synthetic-data notice was removed from the top of the web app.
- The Assistant tab was removed from the web app; the assistant still works from the terminal.
- The app was renamed **Forecast Agent** and the lightning icon removed.

7 Still to check before the demo

- **Run one real question through the assistant.** It has not been called against the live Claude API yet, because no API key was available during development.
- **Add documents to knowledge/** (manuals, tariffs, energy-saving guides) if the assistant should answer questions beyond the forecast and the project's own documentation.
- **Restart the web app** with `streamlit run streamlit_app.py` before presenting.